

LaneBot: Affordable Lane Assistance for Older Vehicles

Project Overview

Duration

24hrs

Tools Used

Figma, Computer Vision, Mobile Development Frameworks, User Research Methods, Prototyping

Project Summary

LaneBot is a mobile-based lane assistance application designed for drivers whose vehicles do not include advanced driver assistance systems (ADAS). The application uses a smartphone camera and real-time lane detection technology to monitor lane positioning and provide timely alerts when a vehicle begins drifting out of its lane.

The goal of LaneBot is to improve road safety by providing an affordable and accessible alternative to expensive built-in lane departure warning systems found in newer vehicles.

Problem Statement

Many older vehicles lack modern safety features such as lane departure warnings and lane-keeping assistance. As a result, drivers may unintentionally drift from their lanes due to fatigue, distraction, poor visibility, or extended highway driving.

Purchasing a newer vehicle with advanced safety features is not financially feasible for many drivers. This creates a significant gap in road safety technology accessibility.

The challenge was to design an intuitive, low-distraction mobile experience that could provide real-time lane assistance without overwhelming or distracting drivers.

Project Goals

The primary goals of LaneBot were:

- Provide affordable lane assistance for vehicles without built-in ADAS.
- Minimize driver distraction while providing safety alerts.

- Deliver real-time lane departure notifications.
- Create a simple and intuitive user experience.
- Ensure accessibility for users of varying technical abilities.
- Maintain a minimal interface suitable for driving environments.

User Research

To understand user needs, research was conducted around common driving challenges and safety concerns.

Key Findings

Driver Fatigue

Drivers frequently experience fatigue during long highway journeys, increasing the likelihood of unintended lane drifting.

Vehicle Limitations

Many drivers own vehicles that lack advanced safety technologies available in newer models.

Distraction Concerns

Drivers prefer solutions that require minimal interaction while driving.

Cost Barriers

Many users seek affordable alternatives to expensive vehicle upgrades.

User Personas

Persona 1: Budget-Conscious Driver

Background

Owns a vehicle manufactured before modern ADAS became common.

Goals

- Improve driving safety.
- Avoid purchasing a new vehicle.

Frustrations

- Limited safety features.
- Expensive upgrade options.

Needs

Affordable safety assistance technology.

Persona 2: Long-Distance Commuter

Background

Frequently drives on highways for extended periods.

Goals

- Reduce risks associated with fatigue.
- Receive timely driving assistance.

Frustrations

- Reduced alertness during long trips.
- Limited awareness of lane drifting.

Needs

Reliable, non-intrusive alerts.

Persona 3: New Driver

Background

Recently obtained a driver's license.

Goals

- Build confidence while driving.
- Receive assistance when making mistakes.

Frustrations

- Anxiety during highway driving.
- Difficulty maintaining lane discipline.

Needs

Simple guidance and reassurance.

User Journey

Scenario: Highway Driving

Step 1

User mounts their smartphone and launches LaneBot.

Step 2

The application calibrates lane detection using the phone camera.

Step 3

The user begins driving.

Step 4

LaneBot continuously monitors lane positioning.

Step 5

The vehicle begins drifting from the center of the lane.

Step 6

LaneBot detects the deviation.

Step 7

An audio and visual alert is triggered.

Step 8

The driver corrects steering and safely returns to the lane.

Why LaneBot Matters

Vehicles on U.S. roads are getting older.

The average age of vehicles in the United States reached **12.8 years in 2025**, the highest on record. This means millions of drivers are operating vehicles that were manufactured before many modern driver-assistance technologies became common.

Many drivers cannot easily afford newer vehicles.

The average new vehicle price exceeded **\$50,000 in late 2025**, and more than **1 in 5 new-car buyers** were taking out loans lasting seven years or longer to afford a purchase.

Most vehicles on the road are older vehicles.

Industry data projects that approximately **70% of vehicles in operation are six years old or older**, meaning a large portion of drivers may not have access to the latest safety technologies.

Safety Gap Between Older and Newer Vehicles

According to the U.S. National Highway Traffic Safety Administration (NHTSA), newer vehicles are significantly safer because of improved crash structures, occupant protection systems, and crash-avoidance technologies.

NHTSA also notes that vehicle fatality rates increase as vehicles age, meaning occupants of older vehicles face greater risk during crashes.

Effectiveness of Lane Departure Systems

This is probably the strongest statistic for your project.

Research from the Insurance Institute for Highway Safety (IIHS) found that vehicles equipped with lane departure warning systems experienced:

- **11% fewer lane-departure-related crashes**
- **21% fewer injury crashes**
- **18% fewer crashes overall**
- **24% fewer injury crashes overall**

Additional IIHS research found lane-departure crash rates were **44% lower** in some vehicles equipped with advanced lane departure prevention systems compared with unequipped vehicles.

User Opportunity Statement

You can include something like:

With the average U.S. vehicle now 12.8 years old and new vehicle prices exceeding \$50,000, millions of drivers continue to operate vehicles without advanced driver assistance systems.

Research shows lane departure warning technologies can reduce relevant crashes by 11% and injury crashes by 21%, creating an opportunity to provide similar safety benefits through a low-cost mobile solution.

A Powerful "Research Insights" Box for Your Portfolio

You could literally place this in a sidebar:

Insight	Evidence
Vehicles are aging	Average vehicle age is 12.8 years in the U.S.
New cars are expensive	Average new vehicle price exceeded \$50,000 in late 2025.
Older vehicles dominate roads	Roughly 70% of vehicles are over six years old.
Lane assistance works	Lane departure warning systems reduce relevant crashes by 11% and injury crashes by 21%.
Newer vehicles are safer	Newer vehicles include crash-avoidance technologies and improved protection systems.

Information Architecture

Application Structure

Home Screen



Camera Calibration



Driving Monitoring Screen



Lane Departure Alerts



Trip Summary



Settings

This structure was designed to minimize navigation complexity and ensure that users could access core functionality with minimal effort.

Design Process

Initial Sketches

Early sketches explored multiple dashboard layouts and alert mechanisms.

The primary focus was reducing visual clutter while maintaining visibility of critical information.

[Insert Sketches]

Low-Fidelity Wireframes

Wireframes were developed to establish:

- Screen hierarchy
- Alert placement
- User flow
- Navigation structure

[Insert Wireframes]

Mid-Fidelity Concepts

Several design iterations were explored to determine the optimal balance between information visibility and distraction reduction.

[Insert Mid-Fidelity Designs]

High-Fidelity Designs

Final interfaces emphasized:

- Simplicity

- High contrast
- Large touch targets
- Clear visual hierarchy

[Insert Figma Screens]

Design Decisions

Audio-First Alerts

Drivers should not be required to continuously monitor the screen.

Audio alerts provide immediate feedback while allowing drivers to remain focused on the road.

Minimal Interface

Only essential information is displayed during driving sessions.

Reducing interface complexity minimizes cognitive load and distraction.

High Contrast Visuals

Driving environments can vary significantly in lighting conditions.

High contrast elements improve visibility during daytime and nighttime usage.

Large Interactive Elements

Large buttons and controls improve usability while reducing interaction difficulty.

Dark Mode Design

A darker interface was selected to reduce glare and visual strain during nighttime driving.

Accessibility Considerations

Accessibility was incorporated throughout the design process.

Features

- High contrast interface
- Readable typography
- Large touch targets
- Audio-based notifications
- Reduced visual clutter
- Color-independent warning indicators

These decisions help ensure usability across a broader range of users.

Safety-Focused UX Decisions

LaneBot operates in a high-risk environment where user attention is critical.

Several UX principles guided the design:

Minimize Cognitive Load

Only critical information is displayed while driving.

Prevent Alert Fatigue

Alerts are triggered only when meaningful lane deviations occur.

Promote Glanceability

Important information can be understood within a fraction of a second.

Reduce Interaction Requirements

The application requires minimal user interaction once driving begins.

Usability Testing

Usability testing was conducted through simulated driving scenarios and user feedback sessions.

Observations

Finding 1

Users preferred audio alerts over vibration-based notifications.

Finding 2

Small visual indicators were often overlooked.

Finding 3

Calibration instructions required additional clarification.

Improvements Implemented

- Increased alert visibility.
- Simplified calibration workflow.
- Enhanced onboarding instructions.
- Reduced unnecessary interface elements.

Edge Cases and Constraints

Several real-world challenges were considered during the design process.

Night Driving

Reduced visibility can impact lane detection accuracy.

Weather Conditions

Rain and fog may affect camera performance.

Faded Lane Markings

Road conditions can reduce detection confidence.

Camera Obstruction

Incorrect phone placement may impact functionality.

Device Performance

Lower-end smartphones may experience processing limitations.

To address these challenges, LaneBot provides user feedback when detection confidence decreases.

Technical Considerations

The design process was closely aligned with technical constraints.

Considerations included:

- Real-time computer vision processing
- Smartphone hardware limitations
- Battery consumption
- Camera quality variations
- Processing latency

Design decisions were made to balance usability and technical feasibility.

Final Solution

LaneBot successfully delivers an affordable lane assistance experience for drivers who lack access to advanced vehicle safety systems.

The final design focuses on:

- Driver safety
- Minimal distraction
- Accessibility
- Simplicity
- Real-time feedback

By combining computer vision technology with user-centered design principles, LaneBot provides a practical and accessible driving safety solution.

Reflection

This project highlighted the importance of designing for safety-critical environments where user attention is limited.

Key lessons learned include:

- Simplicity often improves safety.
- Audio feedback can be more effective than visual feedback in driving scenarios.
- Technical limitations must be considered alongside user needs.
- Designing for edge cases significantly improves overall user experience.

If given additional time, future iterations would explore:

- Voice interaction capabilities
- Personalized alert sensitivity
- Advanced trip analytics
- Integration with wearable devices
- Enhanced environmental awareness features

Key Skills Demonstrated

- User Research
- User Experience Design
- Information Architecture
- Wireframing
- Interaction Design
- Prototyping
- Accessibility Design
- Usability Testing
- Human-Centered Design
- Product Thinking
- Safety-Critical UX Design
- Mobile Application Design
- Computer Vision Product Design

Works Cited

National Highway Traffic Safety Administration. "How Vehicle Safety Has Improved Over the Decades." U.S. Department of Transportation, <https://www.nhtsa.gov/how-vehicle-safety-has-improved-over-decades>.

Reuters. "Americans Keep Vehicles a Record 12.6 Years on Average Despite Easing Supplies." Reuters, 22 May 2024, <https://www.reuters.com/business/autos-transportation/americans-keep-vehicles-record-126-years-average-despite-easing-supplies-sp-says-2024-05-22/>.

S&P Global Mobility. "Average Age of Vehicles in the U.S. Continues to Rise." S&P Global Mobility, 2025, <https://www.spglobal.com/automotive-insights/en/blogs/2025/05/average-age-of-vehicle-in-us>.

U.S. Department of Transportation, Intelligent Transportation Systems Joint Program Office. "Benefits of Lane Departure Warning Systems." ITS Knowledge Resources, <https://www.itskrs.its.dot.gov/2017-b01175>. Accessed 1 June 2026.